Assignment 1

1. Write a Java program to receive any number of command line argument and
a) Display them
b) Display only palindrome string if any.
c) Display count of total arguments.
d) Merge all arguments into a string.
e) Sum of all numbers given in the arguments.

Code:

```
class Cmdlnargs {
	public static void main(String[] args) {
		// display args with their length

		printwithlen(args);

		// display only palindrome string

		onlypalindrome(args);

		// display count of total arguments

		argscount(args);

		// merge all arguments into a string

		mergeallargs(args);

		// display sum of all numbers given in arguments

		addnumargs(args);

	}

	private static void printwithlen(String[] args) {

		System.out.println("1) Print Args With there Length :\n");

		for (int i = 0; i < args.length; i++) {

			String str = args[i];

			System.out.println(str + " " + str.length());

		}

	}

	private static void onlypalindrome(String[] args) {

		System.out.println("2) Print Palindrome Strings : \n");

		for (int i = 0; i < args.length; i++) {

			int left = 0;

			int right = args[i].length() - 1;

			String str = args[i];

			boolean ispalindrome = true;
```

```
                while (right != 0) {
                        if (str.charAt(left) != str.charAt(right)) {
                                ispalindrome = false;
                                break;
                        }
                        left++;
                        right--;
                }
                if (ispalindrome == true) {
                        System.out.println(str);
                }
        }
        System.out.println("---------");
}
private static void argscount(String[] args) {
        System.out.println("3) Print Total Number of Args :\n");
        System.out.println(args.length);
        System.out.println("---------");
}
private static void mergeallargs(String[] args) {
        System.out.println("4) Merge All Args together : \n");
        String str = "";
        for (int i = 0; i < args.length; i++) {
                str = str.concat(args[i]);
        }
        System.out.println(str);
        System.out.println("---------");
}
private static void addnumargs(String[] args) {
        int sum = 0;
        System.out.println("5) Sum of All int Args : \n");
        for (int i = 0; i < args.length; i++) {
                try {
                        sum += Integer.parseInt(args[i]);
```

```
                        } catch (NumberFormatException e) {
                                System.out.println(i + 1 + "st argument isn't an integer");
                        }
                }
                if (sum != 0)
                        System.out.println("Sum of all num args: " + sum);
                System.out.println("---------");
        }
}
```

Output:

```
1) Print Args With there Length :

aba 3
abc 3
dda 3
d 1
1 1
23 2
421 3
121 3
abbba 5
2) Print Palindrome Strings :

aba
d
1
121
abbba
---------
3) Print Total Number of Args :

9
---------
4) Merge All Args together :

abaabcddad123421121abbba
---------
5) Sum of All int Args :

1st argument isn't an integer
2st argument isn't an integer
3st argument isn't an integer
4st argument isn't an integer
9st argument isn't an integer
Sum of all num args: 566
---------
```

2. Using recursion write a program to
a) Display binary conversion of an integer.
b) Verify a number is prime or not.
c) Find Factorial of a number & find nCr = n! / r!(n-r)!

Code (a):

```
class Decitobin {
   public static int binary(int decimal) {
      if(decimal == 0){
         return 0;
      }
      return decimal % 2 + 10 * binary(decimal/2);
   }
   public static void main(String[] args) {
      try {
         int n = Integer.parseInt(args[0]);
         System.out.println(binary(n));
      } catch (Exception e) {
         System.out.println(e);
      }
   }
}
```

Output:

```
PS C:\Users\lenovo\Documents\Java Assignment 1> java Decitobin.java 59
111011
```

Code (b):

```
class Prime {
   public static void main(String[] args) {
      try {
         int inp = Integer.parseInt(args[0]);
         Check c = new Check(inp);
         c.checkprime();
      } catch (Exception e) {
         System.out.println(e);
         System.out.println("Please Enter a valid number");
      }
   }
}
class Check {
```

```
    int num;
    String isPrime(int f) {
        System.out.println("f: " + f);
        System.out.println("num: " + num);
        if (num <= 1) {
            return "is not prime";
        }
        if (f == 1) {
            return "is prime";
        }
        if (num % f == 0 && f < num) {
            return "is not prime";
        }
        return isPrime(f - 1);
    }
    Check(int num) {
        this.num = num;
    }
public void checkprime() {
        System.out.println(isPrime(num));
    }
}
```

Output:

```
PS C:\Users\lenovo\Documents\Java Assignment 1> java Prime 75
is not prime
PS C:\Users\lenovo\Documents\Java Assignment 1>
```

Code (c):

```
class Factorial {
   public static int ncr(int num, int r){
      return fact(num) / fact( num-r) * fact(r);
   }
   public static int fact(int num) {
      if(num ==0){
         return 1;
      }
      return num * fact(num -1);
   }
   public static void main(String[] args) {
      int num = 5;
      int r = 2;
      System.out.println("factorial of " + num + " is " + fact(num));
      System.out.println("nCr of " + num + " & " + r + " is "+ ncr(num, r));
   }
}
```

Output:

```
PS C:\Users\lenovo\Documents\Java Assignment 1> java Factorial
factorial of 5 is 120
nCr of 5 & 2 is 40
```

Assignment 2

1. **Find and replace characters / words (symbols) in a file using RandomAccessFile**

Code:

```
 import java.io.*;
import java.util.Scanner;
 class randomaccessio {
 public static void main(String[] args) throws IOException {
Scanner in = new Scanner(System.in);
System.out.print("Enter a String to match: ");
 String str = in.next();
 System.out.print("Enter String which you want to replace: ");
String replace = in.next();
 RandomAccessFile raf = new RandomAccessFile("R:\\classwork\\text.txt", "rw");
int n = 0;
 while (true) {
long p = raf.getFilePointer();
String s = raf.readLine();
if (s == null) {
    // System.out.println("hii");
    break;
}
String[] arr = s.split(" ");

for (int i = 0; i < arr.length; i++) {
    // System.out.println(arr[i]);
    if (arr[i].equals(str)) {
        raf.seek(p);
        raf.writeBytes(replace);
        n = 1;
        break;
    }
    p += arr[i].length() + 1;
}

}
if (n == 1) {
System.out.println("Replaced Successfully");
```

```
      } else {
        System.out.println("Word Not Found");
      }
      raf.close();
      in.close();
    }
}
```

Output:-

```
Enter a String to match: ramesh
Enter String which you want to replace: rohan
Replaced Successfully
PS C:\Users\raush\OneDrive\Desktop\java\assignment 2>
```

2. Write a contact management program contains:

a. Insert or delete contacts

b. Insert new contacts

c. One name may have more than one contact

d. Modify a particular contact.

e. List of all names who are having maximum number of contacts.

Code:

```
import java.io.*;
import java.util.Scanner;
import java.util.ArrayList;
class contactio {
   //function to show name and contact no.
   public static void nam_no()
     {
      try{
      BufferedReader br = new BufferedReader(new FileReader("contact.txt"));
      String s;
      int i;
      while((s=br.readLine())!= null)
      {
         String[] arr= s.split("##");
         System.out.print(arr[0]+"    ");
         for(i=1;i<arr.length;i++)
```

```
        {
            System.out.print(arr[i]+"  ");
        }

        System.out.println();
    }
    br.close();
}
catch(Exception e)
{
    System.out.print("Error in opening file");
}
}
//function to show name and no of contact
public static void nam_count()
{
    try{
    BufferedReader br = new BufferedReader(new FileReader("contact.txt"));
    String s;
    int i;
    while((s=br.readLine())!=null)
    {
        String[] arr=s.split("##");
        System.out.print(arr[0]+"     ");
        System.out.println(arr.length-1);
    }
}
catch(Exception e)
{
    System.out.print("Error in opening in file");
}
}
//function to add new contact
public static void add_contact()
```

```
{
    try
    {
    BufferedWriter br =new BufferedWriter(new FileWriter("contact.txt",true));
    Scanner in =new Scanner(System.in);
    System.out.print("Enter contact Name :");
    String s=in.next();
    s+="##";
    System.out.print("Enter contact number :");
    String n=in.next();
    s+=n;
    br.write("\n"+s);
    br.close();
    System.out.println("Contact added");
    }
    catch(Exception e)
    {
        e.printStackTrace();
    }
}
//function to delete contact
public static void delete_con()
{
    // Get the name of the contact to be deleted from the user
    Scanner in = new Scanner(System.in);
    System.out.print("Enter the name of the contact to be deleted: ");
    String contactName = in.next();

    // Read the contents of the file into an ArrayList
    ArrayList<String> contacts = new ArrayList<String>();
    try {
        File file = new File("contact.txt");
        Scanner fileScanner = new Scanner(file);
        while (fileScanner.hasNextLine()) {
```

```
            String line = fileScanner.nextLine();
            if (line.startsWith(contactName + "##")) {
                continue;  // skip the line that contains the contact to be deleted
            }
            contacts.add(line);
        }
        fileScanner.close();
    } catch (IOException e) {
        System.out.println("An error occurred while reading the file.");
        e.printStackTrace();
        return;
    }

    // Write the updated list of contacts to the file
    try {
        FileWriter fileWriter = new FileWriter("contact.txt");
        for (String contact : contacts) {
            fileWriter.write(contact + "\n");
        }
        fileWriter.close();
        System.out.println("Contact deleted successfully.");
    } catch (IOException e) {
        System.out.println("An error occurred while writing to the file.");
        e.printStackTrace();
    }
}


public static void main(String[] args)throws IOException {
    System.out.println("Enter Choice");
    System.out.println("1.Show Name and Contact");
    System.out.println("2.Show Name and Number of contacts");
    System.out.println("3.Add contact");
    System.out.println("4.Delete contact");
```

```
        Scanner in=new Scanner(System.in);
        int n= in.nextInt();
        switch(n)
        {
          case 1:
          nam_no();
          break;
          case 2:
          nam_count();
          break;
          case 3:
          add_contact();
          break;
          case 4:
          delete_con();
          break;
          default:
          System.out.println("Enter correct choice");
        }
      }
}
```

Output:-

```
Enter Choice
1.Show Name and Contact
2.Show Name and Number of contacts
3.Add contact
4.Delete contact
1
abc     123   234
def     1111   4343
rohan     3787413
```

3.Create a package consists of utility methods under mypack.util which is capable to do the following things:
a. Display the short form of name
b. Display numeric to words

Code:

```
package mypack;
```

```
public class util
{
   private static final String[] UNITS = {"", "one", "two", "three", "four", "five", "six", "seven", "eight",
"nine"};
   private static final String[] TENS = {"", "ten", "twenty", "thirty", "forty", "fifty", "sixty", "seventy",
"eighty", "ninety"};
   private static final String[] TEENS = {"", "eleven", "twelve", "thirteen", "fourteen", "fifteen",
"sixteen", "seventeen", "eighteen", "nineteen"};
   private static final String HUNDRED = "hundred";
   private static final String THOUSAND = "thousand";
   public String shortNmae(String fname, String lname)
   {
      String sname=fname.charAt(0)+"."+lname;
      return sname;
   }
   public String numToWords(int num)
   {
      if (num == 0) {
         return "zero";
      }
      StringBuilder words = new StringBuilder();
      // handle thousands place
      if (num >= 1000) {
         words.append(numToWords(num / 1000)).append(" ").append(THOUSAND).append(" ");
         num %= 1000;
      }
      // handle hundreds place
      if (num >= 100) {
         words.append(UNITS[num / 100]).append(" ").append(HUNDRED).append(" ");
         num %= 100;
      }
      // handle tens place
      if (num >= 10 && num < 20) {
         words.append(TEENS[num % 10]).append(" ");
         num = 0;
```

```
        } else if (num >= 20) {
            words.append(TENS[num / 10]).append(" ");
            num %= 10;
        }
        // handle units place
        if (num > 0) {
            words.append(UNITS[num]).append(" ");
        }
        return words.toString().trim();
    }
}
```

Output:

```
C:\Users\Gaurav kumar\Desktop\Java> cmd /C ""C:\Program Files\Java\jdk-19\bin\java.exe" --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\U
sers\Gaurav kumar\AppData\Roaming\Code\User\workspaceStorage\154945324807785f0743b19447e884a0\redhat.java\jdt_ws\Java_5745b8e5\bin" Project.mypackage.Main
 "
Short name: J.D.
Number in words: one hundred twenty three million four hundred fifty six thousand seven hundred eighty nine
```

4. Create one subpackage file under mypack.file which contains: a. Find and replace substring in a file

Code:

```
package mypack.file;

import java.io.*;

public class file {

    public String findRplace(String find,String replace,String address) throws
IOException

    {

            RandomAccessFile raf = new RandomAccessFile(address, "rw");

            int n = 0;

            while (true) {

                long p = raf.getFilePointer();
```

```
            String s = raf.readLine();
            if (s == null) {
                // System.out.println("hii");
                break;
            }
            String[] arr = s.split(" ");

            for (int i = 0; i < arr.length; i++) {
                // System.out.println(arr[i]);
                if (arr[i].equals(find)) {
                    raf.seek(p);
                    raf.writeBytes(replace);
                    n = 1;
                    break;
                }
                p += arr[i].length() + 1;
            }
        }
        String str;
        if (n == 1) {
          str =  "Replaced Successfully";
        } else {
           str="Word Not Found";
        }
        raf.close();
        return str;
    }
```

```
}
```

5. Write two threads points table of N where N is given by the user.

```
 import java.util.Scanner;
class PrintTable implements Runnable {
   static int print_num=10;
   static int num =1;
   private int n;
   int remainder;
   static Object obj = new Object();
   PrintTable(int remainder,int n)
   {
      this.remainder= remainder;
      this.n=n;
   }
   @Override
   public void run()
   {
      while(num < print_num ){
         synchronized (obj){
            if(num%2 !=remainder){
               try {
                  obj.wait();
               } catch (Exception e) {
                  e.printStackTrace();
               }
            }
```

```
            System.out.println(Thread.currentThread().getName()+" "+num*n);
            num++;
            obj.notifyAll();
        }
    }
    print_num--;
  }
}
public class PrintTableN {
  public static void main(String[] args) {
    System.out.print("Enter the First Number of the Table =");
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    PrintTable Run1 = new PrintTable(1,n);
    PrintTable Run2 = new PrintTable(0,n);
    Thread t1 = new Thread(Run1,"Thread1");
    Thread t2 = new Thread(Run2,"Thread2");
    t1.start();
    t2.start();
  }
}
```

Output:-

```
Enter the First Number of the Table =4
Thread1 4
Thread2 8
Thread1 12
Thread2 16
Thread1 20
Thread2 24
Thread1 28
Thread2 32
Thread1 36
Thread2 40
```

6. Design two threads which print a series of numbers upper and lower limits taken from users before thread generation without any repetition.

```
import java.util.Scanner;

class PrintNumbers implements Runnable {

    private int id;

    private int lower;

    private int higher;

    private static Object lock = new Object();

    private static int current = 1;



    public PrintNumbers(int id, int lower, int higher) {

        this.id = id;

        this.lower = lower;

        this.higher = higher;

        current = lower;

    }

    @Override

    public void run() {

        synchronized (lock) {

            while (current <= higher) {

                if (current % 2 == id) {
```

```
                System.out.println(Thread.currentThread().getName() + " prints " +
current);

                current++;

                lock.notifyAll();

            } else {

                try {

                    lock.wait();

                } catch (InterruptedException e) {

                    e.printStackTrace();

                }

            }

        }

        lock.notifyAll();

    }

}}

public class thread6 {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the lower limit: ");

        int lower = scanner.nextInt();

        System.out.print("Enter the higher limit: ");

        int higher = scanner.nextInt();

        PrintNumbers evenPrinter = new PrintNumbers(0, lower, higher);

        PrintNumbers oddPrinter = new PrintNumbers(1, lower, higher);

        Thread evenThread = new Thread(evenPrinter, "Even Thread");

        Thread oddThread = new Thread(oddPrinter, "Odd Thread");

        evenThread.start();
```

```
        oddThread.start();

    }

}
```

Output:-

```
Enter the lower limit: 2
Enter the higher limit: 8
Even Thread prints 2
Odd Thread prints 3
Even Thread prints 4
Odd Thread prints 5
Even Thread prints 6
Odd Thread prints 7
Even Thread prints 8
```

7.Write a Java program to create two threads to display a message in proper way to consol.

```
public class thread7 extends Thread {

    private String message;

    private int delay;

    public thread7(String message, int delay) {

        this.message = message;

        this.delay = delay;

    }

    public void run() {

        int i=0;

        try {

            while (i<10) {

                System.out.println(message);

                Thread.sleep(delay);

                i++;

            }

        } catch (InterruptedException e) {
```

```
            // Thread interrupted
        }
    }

    public static void main(String[] args) {
        thread7 thread1 = new thread7("Hello from Thread 1!", 1000);
        thread7 thread2 = new thread7("Hello from Thread 2!", 2000);
        thread1.start();
        thread2.start();
    }
}
```

Output:-

```
Hello from Thread 1!
Hello from Thread 2!
Hello from Thread 1!
Hello from Thread 2!
Hello from Thread 1!
Hello from Thread 1!
Hello from Thread 2!
Hello from Thread 1!
Hello from Thread 1!
Hello from Thread 2!
Hello from Thread 1!
Hello from Thread 1!
Hello from Thread 2!
Hello from Thread 1!
Hello from Thread 1!
Hello from Thread 2!
Hello from Thread 2!
Hello from Thread 2!
Hello from Thread 2!
Hello from Thread 2!
```

Assignment 3

1. **Using prepared statement insert 1000 employee(id, name, dob) into the databse.**

Code:

```
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.Connection;
import java.util.Properties;
import java.util.Scanner;
class assign1{
   public static void main(String[] args) throws Exception {
      //connect to databse
      String constr = "jdbc:mysql://localhost:3306/raushan";
      Connection con = null;
      PreparedStatement stmt = null;
      Properties prop = new Properties();
      prop.put("user", "root");
      prop.put("password","12345");
      try {
         con = DriverManager.getConnection(constr, prop);
         //inserting data
         for (int i = 0; i < n; i++) {
            stmt = con.prepareStatement("Insert into employee(emp_id,name,dob)
values(?,?,?)");
            stmt.setInt(1, id);
            stmt.setString(2,”ram”);
            stmt.setString(3,”10/10/2000”);
            stmt.executeUpdate();
         }
      }  System.out.println("Data Inserted Successful");
catch(Exception e) {
         e.printStackTrace();
      } finally {
         if (stmt != null) {
            stmt.close();
         }
         if (con != null) {
            con.close();
         }
      }
   }
}
```

Output:-

```
PS C:\Users\raush\OneDrive\Desktop\java\assignment 3> java -cp "r:/jars/con.jar;" assign1
Data Inserted Successful
```

2. **Using ResultSet insert five rows in employee(id,name,dob). Display the table records before insertion & after each insertion using the same ResultSet used for insertion.**

Code:-

```
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Properties;
import java.sql.DriverManager;

class assign2
{
   public static void main(String[] args) {
      //connect to database
      Connection con=null;
      PreparedStatement stmt=null;
      String constr="jdbc:mysql://localhost:3306/raushan";
      Properties pro=new Properties();
      pro.put("user","root");
      pro.put("password","12345");
      try{

         con = DriverManager.getConnection(constr,pro);
         //Before Insertion
          stmt =con.prepareStatement("select * from employee",3,2);
         ResultSet rs = stmt.executeQuery();
         System.out.println("Before Insert");
         while(rs.next())
         {
            System.out.print(rs.getInt(1)+"  ");
            System.out.print(rs.getString(2)+" ");
            System.out.println(rs.getString(3));
         }
        // System.out.println("Enter employee id,Nameand DOB");
         for(int i=10;i<16;i++)
         {
            stmt=con.prepareStatement("insert into employee values(?,?,?)",3,2);
            stmt.setInt(1,i);
            stmt.setString(2,"ram" );
            stmt.setString(3, "1/1/111");
            stmt.executeUpdate();
         }
         //After insertion
         System.out.println("After insert");
         stmt =con.prepareStatement("select * from employee",3,2);
         rs = stmt.executeQuery();
```

```
            while(rs.next())
            {
                System.out.print(rs.getInt(1)+"  ");
                System.out.print(rs.getString(2)+" ");
                System.out.println(rs.getString(3));
            }
        }
        catch(Exception e)
        {
            e.printStackTrace();
        }
        finally
        {

        }
    }
}
```

Output:-

```
Before Insert
1  mohan 10/2/2000
2  sohan 14/8/2000
3  sita 19/9/2000
4  gita 12/3/2000
After insert
1  mohan 10/2/2000
2  sohan 14/8/2000
3  sita 19/9/2000
4  gita 12/3/2000
10  ram 1/1/111
11  ram 1/1/111
12  ram 1/1/111
13  ram 1/1/111
14  ram 1/1/111
15  ram 1/1/111
```

3. **Using callabel statement call the stored procedure "proc 1()" which selecs two tables of unknown structure . Display those tables using the callable statement object.**

Code:-

```
import java.sql.*;
import java.util.Properties;

public class assign3 {
    public static void main(String[] args) {
        Connection con = null;
        CallableStatement stmt = null;
        try {
            // Set up the database connection properties
            String url = "jdbc:mysql://localhost:3306/raushan";
```

```
    String username = "root";
    String password = "12345";
    Properties props = new Properties();
    props.put("user", username);
    props.put("password", password);

    // Connect to the database and call the stored procedure
    con = DriverManager.getConnection(url, props);
    stmt = con.prepareCall("{call proc1()}");
    boolean hasResult = stmt.execute();

    // Process the result sets
    while (hasResult) {
        ResultSet rs = stmt.getResultSet();
        ResultSetMetaData rsmd = rs.getMetaData();
        int numCols = rsmd.getColumnCount();

        // Print the column headers
        for (int i = 1; i <= numCols; i++) {
            System.out.print(rsmd.getColumnLabel(i) + "\t");
        }
        System.out.println();

        // Print the rows of the result set
        while (rs.next()) {
            for (int i = 1; i <= numCols; i++) {
                System.out.print(rs.getString(i) + "\t");
            }
            System.out.println();
        }

        // Check if there are more result sets
        hasResult = stmt.getMoreResults();
    }
} catch (SQLException e) {
    e.printStackTrace();
} finally {
    // Close the statement and connection objects
    try {
        if (stmt != null) {
            stmt.close();
        }
        if (con != null) {
            con.close();
        }
    } catch (SQLException e) {
        e.printStackTrace();
```

```
        }
      }
    }
  }
```

Stored Procedure:

```
CREATE DEFINER=`root`@`localhost` PROCEDURE `proc1`()
BEGIN
    SELECT * from employee;
    SELECT * FROM student;
END
```

Output:-

emp_id	Name	DOB
1	mohan	10/2/2000
2	sohan	14/8/2000
3	sita	19/9/2000
4	gita	12/3/2000
10	ram	1/1/111
11	ram	1/1/111
12	ram	1/1/111
13	ram	1/1/111
14	ram	1/1/111
15	ram	1/1/111
roll	name	
100	ram	

4. **Call the stored procedure getResultSet(id (in), str (out)) which Selects in all sales items seperated by comma and at the last name of the id, and the tables are in use; Salesman(id, name) Sale (id, item).**

Code:-

```
import java.sql.*;
import java.util.Scanner;

public class assign4 {
   public static void main(String[] args) {
      try {


         // Connect to the database
         String url = "jdbc:mysql://localhost:3306/raushan";
         String user = "root";
         String password = "12345";
         Connection conn = DriverManager.getConnection(url, user, password);

         // Prepare the stored procedure call
         CallableStatement stmt = conn.prepareCall("{call getResult(?, ?)}");
```

```
        // Set the input parameter for the stored procedure
        Scanner in= new Scanner(System.in);
        System.out.print("Enter Id: ");
        int id = in.nextInt();
        stmt.setInt(1, id);
        // Register the output parameter for the stored procedure
        stmt.registerOutParameter(2, Types.VARCHAR);

        // Execute the stored procedure
                stmt.execute();
           // Get the output parameter value from the stored procedure
        String str = stmt.getString(2);
        if(str==null)
        {
           System.out.println("Id not found");
        }
        else
        {
        System.out.println("Output parameter value: " + str);
        }
        // Close the statement and connection
        stmt.close();
        conn.close();
     } catch (Exception e) {
        e.printStackTrace();
     }
   } }
```

Stored Procedure:-

```
CREATE DEFINER=`root`@`localhost` PROCEDURE `getResult`(IN id INT, OUT str VARCHAR(255))
BEGIN
    DECLARE item_names VARCHAR(255);
    DECLARE seller_name VARCHAR(255);

    SELECT GROUP_CONCAT(item_name SEPARATOR ', ') INTO item_names
    FROM sales
    WHERE salesman_id = id;

    SELECT name INTO seller_name
    FROM salesman
    WHERE salesman_id = id;

    SET str = CONCAT(item_names, ', ', seller_name);
END
```

Output:-

```
Enter Id: 3
Output parameter value: Product d, Product c, Product f, Mohan
```

Assignment 4

1. **Write a Servlet program to connect to a database and generate the page contains Tables combo, Fields combo and data area (html table):**
   Table combo contains all table names of the connected database.
   On selection of table combo, Field combo fills with all fields of the selected table.
   On selection of field combo all data of selected field are to be displayed on the data area. If we select all fields from Field combo then respective table data are to be displayed.

Code:

```
package servlets;

import java.sql.*;
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class Test extends HttpServlet {

// declare the database connection parameters

 String databaseURL = "jdbc:mysql://localhost:3306/gauravdb";
 String username = "root";
 String password = "0000";
 Connection conn = null;
 Statement stmt = null;
 ResultSet rs = null;

 // override the init() method to initialize the database connection

 public void init() throws ServletException {
try {
        Class.forName("com.mysql.jdbc.Driver");
        conn = DriverManager.getConnection(databaseURL, username, password);
        stmt = conn.createStatement();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

 // override the doGet() method to handle the GET request
 protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException,
IOException{
   String selectedTableName = req.getParameter("table-name");
   String[] selectedFields = req.getParameterValues("field-names");
```

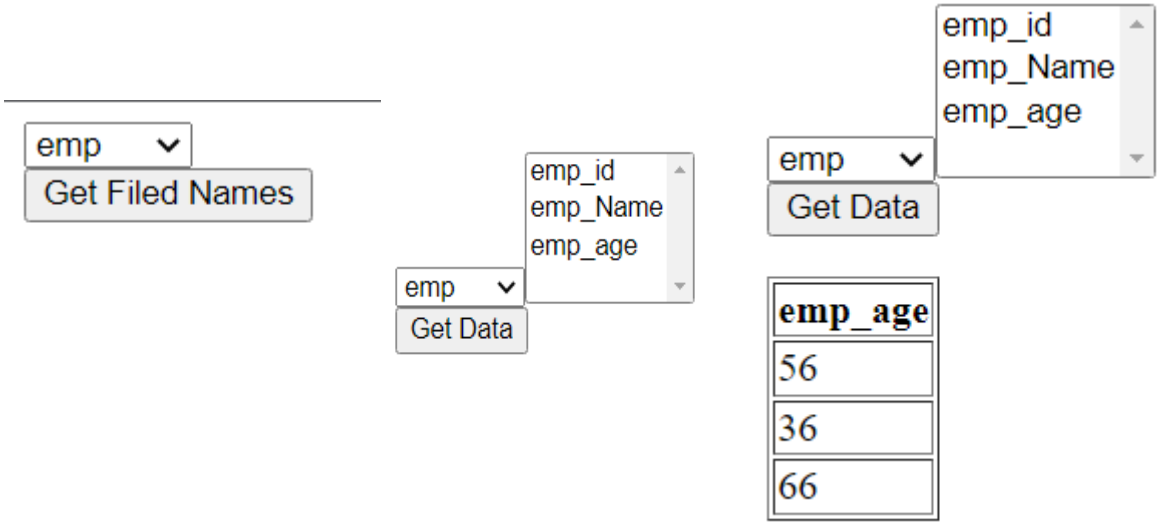
```
resp.setContentType("text/html");
PrintWriter out = resp.getWriter();
out.print("<html><head>");
out.print("<title>Databases Viewer</title>");
out.print("</head>");
out.print("<body></center>");
   try(Statement stm = conn.createStatement()) {
ResultSet rs = stm.executeQuery("SHOW TABLES");
   out.print("<form>");
   out.print("<select name='table-name'>");
   while (rs.next()) {
      String tableName = rs.getString(1);
 if (tableName.equals(selectedTableName)) {
         out.print("<option value='"+tableName+"' selected>"+tableName+"</option>");
      } else{
         out.print("<option value='"+tableName+"'>"+tableName+"</option>");
      }
   }
   out.print("</select>");
   if (selectedTableName == null) {
      out.print("<br><input type='submit' value='Get Filed Names'>");
      out.print("</form>");
      return;
   }
   out.print("<select name='field-names' multiple>");
   rs = stm.executeQuery("DESC "+selectedTableName);
   while (rs.next()) {
      String filedName = rs.getString(1);
      out.print("<option value='"+filedName+"'>"+filedName+"</option>");
   }
   out.print("</select>");
    out.print("<br><input type='submit' value='Get Data'>");
  out.print("</form>");
   if (selectedFields == null || selectedFields.length == 0) return;
   StringBuilder columnNames = new StringBuilder();
   if (selectedFields.length == 1) {
      columnNames.append(selectedFields[0]);
   } else {
      for (int i = 0; i < selectedFields.length; i++) {
         if (i == selectedFields.length-1) {
            columnNames.append(selectedFields[i]);
            break;
         }
         columnNames.append(selectedFields[i]+", ");
      }
   }
         rs = stm.executeQuery("SELECT "+columnNames.toString()+" FROM " +
selectedTableName);
out.print("<table border='1'>");
   ResultSetMetaData meta = rs.getMetaData();
```

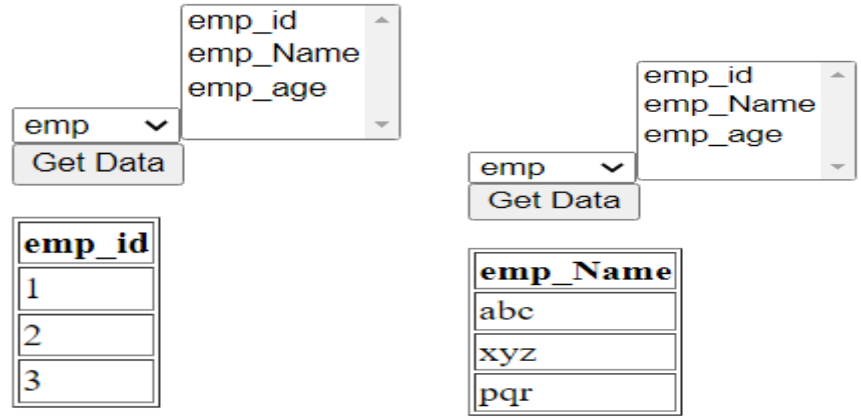
```
        int colCount = meta.getColumnCount();
        out.print("<tr>");
        for (int i = 1; i <= colCount; i++) {
           out.print("<th>"+meta.getColumnLabel(i)+"</th>");
        }
 out.print("</tr>");
        while (rs.next()) {
           out.print("<tr>");          for (int i = 1; i <= colCount; i++) {
              out.print("<td>" + rs.getString(i)+"</td>");
           }
out.print("</tr>");
        }
        out.print("</table>");
    } catch (SQLException e) {
        out.print("Error occurred: " +e.getMessage());
    }
 out.print("</center></body></html>");
}
@Override
protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException,
IOException {
    doGet(req, resp);
}
}
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi=http://www.w3.org/2001/XMLSchema-instance
        xmlns=http://java.sun.com/xml/ns/javaee
        xsi:schemaLocation=http://java.sun.com/xml/ns/javaee
                    http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd
        id="WebApp_ID" version="2.5">
  <display-name>jdbc_servlet</display-name>
  <welcome-file-list>
    <welcome-file>index.html</welcome-file>
  </welcome-file-list>
  <servlet>
    <description></description>
    <servlet-name>Test</servlet-name>
    <servlet-class>servlets.Test</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>Test</servlet-name>
    <url-pattern>/Test</url-pattern>
  </servlet-mapping>
 </web-app>
```

Output :





2. **Write a Servlet program to demonstrate initial parameter access from a servlet page.**

Code:

```
package servlets;


import jakarta.servlet.ServletConfig;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;


public class app extends HttpServlet {
   public void doGet(HttpServletRequest req,HttpServletResponse res) throws
ServletException,IOException
```

```
  {
    res.setContentType("text/html");
    PrintWriter out=res.getWriter();
    ServletConfig config = this.getServletConfig();
    String value1=config.getInitParameter("parameter_1");
    String value2=config.getInitParameter("parameter_2");
    out.println("<h3>First intial parameter: "+"<br>"+value1+"</h3>");
    out.println("<h3>Second intial parameter: "+"<br>"+value2+"</h3>");
  }
}
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi=http://www.w3.org/2001/XMLSchema-instance
      xmlns=http://java.sun.com/xml/ns/javaee
      xsi:schemaLocation=http://java.sun.com/xml/ns/javaee
                    http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd
      id="WebApp_ID" version="2.5">
 <display-name>init_para</display-name>
 <welcome-file-list>
  <welcome-file>index.html</welcome-file>
 </welcome-file-list>
 <servlet>
  <description></description>
  <servlet-name>app</servlet-name>
  <servlet-class>servlets.app</servlet-class>
  <init-param>
    <param-name>parameter_1</param-name>
    <param-value> 1st initial parameter</param-value>
  </init-param>
  <init-param>
    <param-name>parameter_2</param-name>
    <param-value> 2nd initial parameter</param-value>
  </init-param>
 </servlet>
 <servlet-mapping>
  <servlet-name>app</servlet-name>
  <url-pattern>/app</url-pattern>
 </servlet-mapping>
</web-app>
```

Output:

First intial parameter:
This is first initial parameter

Second intial parameter:
This is second initial parameter

3. **Write a Servlet program to demonstrate parameter (with/ without names, single valued/multiple valued) access from a servlet page.**

Code:

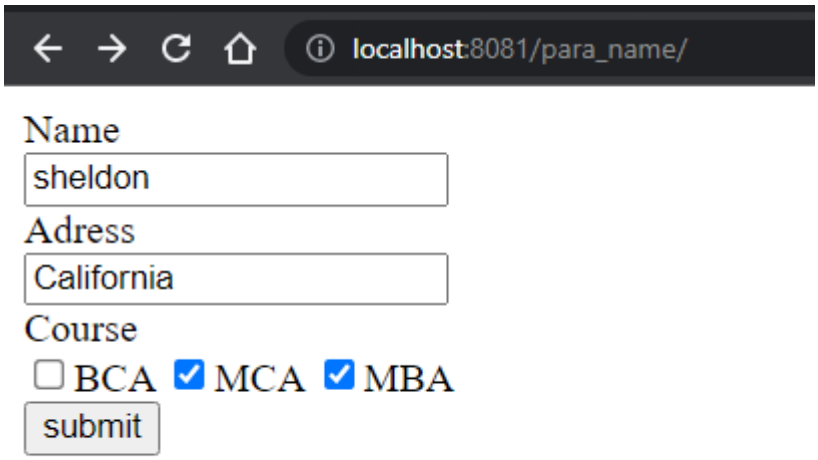
```
package servlets;

import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.util.Enumeration;
import java.io.IOException;
import java.io.PrintWriter;
public class app extends HttpServlet {
   public void doGet(HttpServletRequest req,HttpServletResponse res) throws
ServletException,IOException

   {

      res.setContentType("text/html");
      PrintWriter out=res.getWriter();
      //Accessing parameter through single value names
      String name= req.getParameter("name");
      String address= req.getParameter("add");
      out.println("<h3>Accessing single valued parameter through name</h3>");
      out.println(name+"<BR>");
      out.println(address+"<BR>");
      //Accessing multi valued parameter with name
      out.println("<h3>Accessing multivalued valued parameter through name</h3>");
      String[] course=req.getParameterValues("chk");
      for(int i=0;i<course.length;i++)
      {

        out.println(course[i]+"<BR>");       }

     //Acessing parameter with enumeration
 out.println("<h3>Accessing  parameter through Enumeration</h3>");
      Enumeration allnames=req.getParameterNames();
      while(allnames.hasMoreElements())
```

```
    {
      String pname=(String)allnames.nextElement();
      String[] values=req.getParameterValues(pname);
      for(String val: values)
      {
        out.println(val+"<BR>");
      }
    }
  }
  }
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi=http://www.w3.org/2001/XMLSchema-instance
      xmlns=http://java.sun.com/xml/ns/javaee
      xsi:schemaLocation=http://java.sun.com/xml/ns/javaee
                    http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd
      id="WebApp_ID" version="2.5">
 <display-name>Para_name</display-name>
 <welcome-file-list>
  <welcome-file>index.html</welcome-file>
 </welcome-file-list>
 <servlet>
  <description></description>
  <servlet-name>app</servlet-name>
  <servlet-class>servlets.app</servlet-class>
 </servlet>
 <servlet-mapping>
  <servlet-name>app</servlet-name>
  <url-pattern>/app</url-pattern>
 </servlet-mapping>
</web-app>
```

Output:



Accessing single valued parameter through name

sheldon
California

Accessing multivalued valued parameter through name

MCA
MBA

Accessing parameter through Enumeration

sheldon
California
MCA
MBA

4. **Create one servlet page contains one text box to receive user name and submit button. On submit the page welcomes the user if the user visits the page for the first time else print "hello <user> you are visiting this site for N times". If no user typed then print "Select User"**

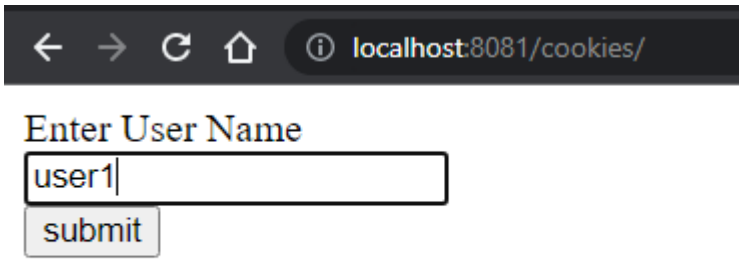
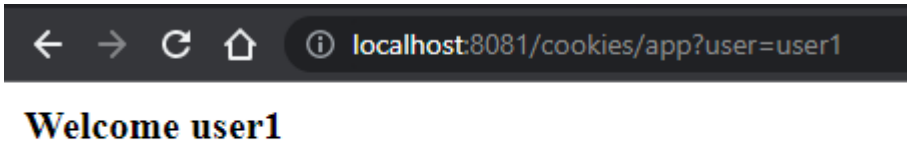
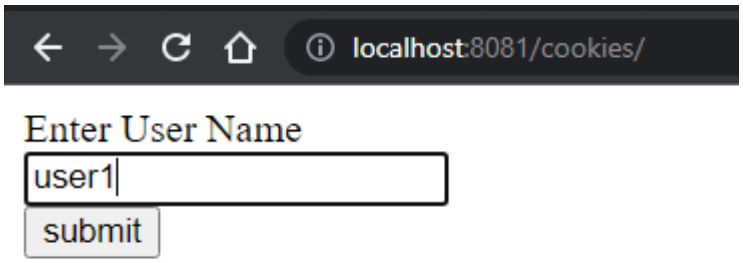
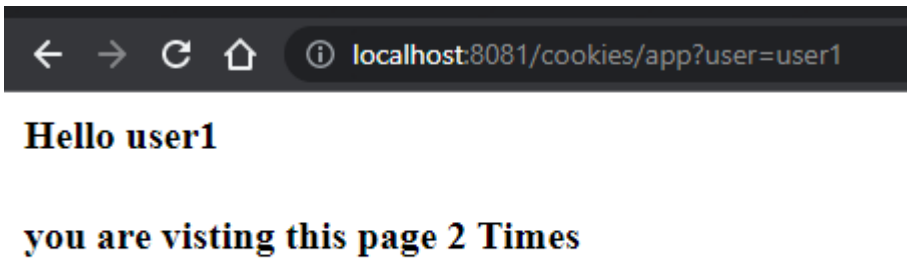
Code:

```
package servlets;


import jakarta.servlet.ServletException;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;


public class app extends HttpServlet {

   @Override
   protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException,
IOException {
      resp.setContentType("text/html");
      PrintWriter out = resp.getWriter();
      String user = req.getParameter("user");
      int i = 0;
      if (user.equals("")) {
         out.print("<h2>Select User</h2>");
      } else {
```

```
        Cookie[] cks = req.getCookies();
        if (cks != null) {
            for (Cookie ck : cks) {
                if (ck.getName().equals(user)) {
                    out.println("<h4>Hello " + user + "</h4>");
                    ck.setValue(String.valueOf(Integer.parseInt(ck.getValue()) + 1));
                    out.print("<h4> you are visiting this page " + ck.getValue() + " Times</h4>");
                    out.print("<a href='/clear-cookies'>Clear Cookies</a>");
                    resp.addCookie(ck);
                    i = 1;
                }
            }

            if (i == 0) {
                out.print("<h4>Welcome " + user + "</h4>");
                Cookie ck1 = new Cookie(user, "1");
                resp.addCookie(ck1);
            }
        } else {
            out.print("<h4>Welcome " + user + "</h4>");
            Cookie ck1 = new Cookie(user, "1");
            resp.addCookie(ck1);
        }
    }

  }


  @Override
  protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {
      doGet(req, resp);
  }

}
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi=http://www.w3.org/2001/XMLSchema-instance
      xmlns=http://java.sun.com/xml/ns/javaee
      xsi:schemaLocation=http://java.sun.com/xml/ns/javaee
                    http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd
      id="WebApp_ID" version="2.5">
 <display-name>Cookies</display-name>
 <welcome-file-list>
  <welcome-file>index.html</welcome-file>
 </welcome-file-list>
 <servlet>
  <description></description>
  <servlet-name>app</servlet-name>
  <servlet-class>servlets.app</servlet-class>
```

```
  </servlet>
  <servlet-mapping>
    <servlet-name>app</servlet-name>
    <url-pattern>/app</url-pattern>
  </servlet-mapping>

</web-app>
```

Output